

Wazuh Agent

- Wazuh is an open-source security monitoring platform (SIEM + XDR) . It watches systems and says: Something suspicious happened!
- It is used for: Threat detection , Log analysis , File integrity monitoring , Intrusion detection ,Endpoint monitoring.
- Is Wazuh Similar to Kibana?
Kibana is only a dashboard tool used for visualizing logs . Wazuh is a complete security platform, not just a dashboard.
But internally Wazuh actually uses Elasticsearch for storing data and a dashboard similar to Kibana
- Elastic uses Fleet for centralized management, whereas Wazuh uses a manager that performs log analysis and rule-based intrusion detection.
- Wazuh has built-in HIDS (Host Intrusion Detection System).
It can detect: Rootkits , Suspicious file changes, Malware behavior , Unauthorized access
Example:If someone changes a critical file like: /etc/passwd
Wazuh will immediately generate an alert.
This happens because Wazuh uses:File Integrity Monitoring (FIM) , Rootcheck , Syscheck

Hence we can say that Wazuh focuses more on endpoint intrusion detection, whereas Elastic focuses more on log analytics and SIEM visibility.

➤ WHT HAPPENS WHEN AGENT IS INSTALLED ON ENDPOINT :

1. A background service starts running.
2. The agent registers with the SIEM server.
3. It begins collecting logs and system telemetry.
4. The data is securely transmitted to the central server.

Example Linux services: elastic-agent.service
wazuh-agent.service

These services continuously monitor the system.

➤ What security risks exist if agents are compromised?

If an attacker compromises an agent:

- they may tamper with logs
- disable monitoring
- send false telemetry
- hide malicious activity

To prevent this, SIEM systems use:

- TLS encryption
- authentication keys
- secure agent registration
- access control policies

➤ Difference between **agent-based monitoring** and **agentless monitoring**?

Answer: **Agent-based monitoring**

Requires software installed on endpoints , Provides deep system visibility , More accurate security monitoring

Agentless monitoring : Collects data remotely , Uses protocols like SSH, API, SNMP , Easier to deploy but provides limited visibility

1. Wazuh Architecture : 4 components

Component	Purpose	Computer
Wazuh Agent	Installed on endpoint (Linux/Windows/Mac)	Wazuh Agent
Wazuh Manager	Processes logs and rules	Wazuh Manager
Wazuh Indexer	Stores security data	Wazuh Indexer
Wazuh Dashboard	Visual interface	Dashboard

- **Wazuh Agent** : It acts like a **sensor** that collects security data from the system

What the Agent Actually Does : The agent continuously monitors the endpoint and collects:

- System logs • Login activity • Running processes • File changes • Network connections

- **Wazuh Manager** (The Brain) : is the central analysis server. It is basically the core detection engine.

All agents send their logs to the manager. Now the manager

1. Receives logs from agents
2. Analyzes the logs
3. Applies security rules
4. Generates alerts

- **Wazuh Indexer** (Data Storage Engine) : stores all security data. It is based on Elasticsearch

Important because : fast searching , log analysis , historical investigation

Indexer Stores : Security events , Agent logs , Alerts , system metrics

- **Wazuh Dashboard** (Visualization Layer) : user interface. It shows all the security information visually.

2. D/W

Elastic	Wazuh
Elastic Agent installed on endpoint	Wazuh Agent installed on endpoint
Fleet server manages agents	Wazuh Manager manages agents
Data stored in Elasticsearch	Data stored in Wazuh Indexer
Visualized in Kibana	Visualized in Wazuh Dashboard

Security	API key / enrollment token	Agent key + authentication
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Elastic agent communicates with the Fleet server, whereas Wazuh agent communicates with the Wazuh manager.

Both collect endpoint telemetry like logs, processes, and security events, but Wazuh also focuses heavily on host intrusion detection (HIDS) features such as rootkit detection and file integrity monitoring.

Rootkit Detection :

What is a Rootkit? - A rootkit is a type of malware that hides inside a system and gives attackers secret administrator (root) access.

It is dangerous because it tries to hide itself from the operating system and security tools.

Example : An attacker gains access to a Linux server and installs a rootkit.

The rootkit might: Hide attacker processes , Hide malicious files , Hide backdoor connections

So even if the administrator runs commands like:

```
ps  
netstat
```

the malicious activity might not appear.

- Security tools like Wazuh run rootkit detection scans.

They check for: suspicious binaries ,hidden processes ,modified system commands ,known rootkit signatures . If something suspicious is detected, an alert is generated.

- **File integrity monitoring** tracks changes in critical system files by comparing hash values, and it generates alerts if unauthorized modifications occur.

3. Services Running After Agent Installation:

When you install an agent on Linux, background services start running.

- For Elastic Agent - Command to check: `sudo systemctl status elastic-agent`

Main service: `elastic-agent.service`

Internal components running inside Elastic Agent

Filebeat module : Collects system logs

Metricbeat module : Collects system metrics , CPU usage , memory usage , disk usage

System monitoring module : tracks system processes , tracks login events

Functions: Collect logs , Monitor system activity , Send data to Elastic server

- For Wazuh Agent - Command: `sudo systemctl status wazuh-agent`

Main service: `wazuh-agent.service`

Sub-modules inside the agent: Log collector , File integrity monitoring ,Rootcheck , Syscheck , Command execution module.

4. Authentication Mechanism (Security Perspective)

- Elastic Agent Authentication

Uses:

- Enrollment Token
- API Key
- TLS encryption

Flow:

Agent → Fleet Server → Elasticsearch

- Wazuh Agent Authentication

Uses:

- Agent key
- Manager authentication
- TLS encryption

Flow:

Agent → Wazuh Manager → Indexer / Dashboard

Each agent gets a unique key when registered.

- **Elastic Authentication**

Flow:

Agent install :



Enrollment Token used



Fleet validates token



Agent registered



API Key generated



Agent communicates using API Key + TLS

Step 1 — Enrollment Token Creation : In Kibana → Fleet, a Enrollment Token is generated.

Step 2 — Agent Installation

When you install the agent you run something like

```
sudo ./elastic-agent install \  
--url=https://fleet-server:8220 \  
--enrollment-token=XXXX
```

Here the agent sends: Enrollment Token → Fleet Server then Fleet verifies the token.

Step 3 — Agent Registration - If the token is valid:

- Fleet registers the agent
- Assigns an Agent ID
- Generates an API Key

Step 4 — API Key Authentication - After registration, the API key is used for continuous communication. Why? - Because the enrollment token is only for first registration.

Now communication becomes: Agent → Fleet Server using API Key , now API key acts like a permanent credential.

Step 5 — TLS Encryption - All communication uses TLS (Transport Layer Security).

TLS provides: Encryption , Authentication , Data integrity

This prevents: Man-in-the-middle attacks , data interception

- **Wazuh Agent Authentication.**

Step 1 — Agent Installation - Agent is installed on endpoint. (Linux, Mac , Windows)

Step 2 — Agent Registration - The agent must register with Wazuh Manager.

During registration: Agent → Manager

Manager assigns: Agent ID , Agent Key

Step 3 — Agent Key Authentication - The Agent Key is stored locally and used for authentication.

It is stored in: /var/ossec/etc/client.keys

This key allows secure communication between: Agent ⇔ Manager

Step 4 — Secure Communication - Like Elastic, Wazuh also uses TLS encryption.

So communication becomes: Encrypted communication

Hence Authentication Is Important

Without authentication, an attacker could:

- send fake logs
- inject malicious events
- impersonate endpoints

With Wazuh authentication:

- every agent has a unique ID
- every agent has a unique key
- communication is encrypted

This ensures secure endpoint monitoring.

Elastic

Temporary Entry Pass → Enrollment Token

Permanent ID Card → API Key

Security Camera → TLS encryption

Wazuh

Register at reception → Agent registration

Permanent building ID → Agent key

Secure communication → TLS encryption

5. Can the same agent be used on another system? Or One Agent Be Used On Multiple Systems

No ,Reason:

- Each agent generates a unique agent ID
- It is registered to a specific endpoint
- Authentication key is unique • unique authentication key • unique hostname

Example:

System 1 → Agent ID 001

System 2 → Agent ID 002

System 3 → Agent ID 003

The server tracks each system separately. Even if you copy the files, the manager will detect mismatch , as the hostname will change , system fingerprint will change

Hence, From a security perspective, deploying agents across multiple operating systems like Windows, Linux, and Mac helps achieve better endpoint visibility because each agent collects system-specific logs and telemetry from that operating system.

In Elastic ,

Elastic Agent uses:

- Enrollment Token (first registration)
- API Key (for continuous authentication)

When the agent registers with Fleet:

- Fleet assigns a unique agent identity
- The agent becomes linked to that endpoint

Copying the agent files to another machine will cause authentication issues.

6. If the WASU agent is deployed in Windows, can it be visualized in Kibana?

Yes , because Wazuh sends the processed events to Elasticsearch, and Kibana is used to visualize the data stored in Elasticsearch.

Process: The agent collects endpoint logs from Windows, sends them to the Wazuh manager for analysis, and the processed security events are indexed(stored) in Elasticsearch, which can then be visualized in Kibana dashboards.